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Signal alignment for cross-datasets in P300 brain-computer interfaces

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Abstract

Objective. Transfer learning has become an important issue in the brain-computer interface (BCI) field, and studies on subject-to-subject transfer within the same dataset have been performed. However, few studies have been performed on dataset-to-dataset transfer, including paradigm-to-paradigm transfer. In this study, we propose a signal alignment (SA) for P300 event-related potential (ERP) signals that is intuitive, simple, computationally less expensive, and can be used for cross-dataset transfer learning. *Approach.* We proposed a linear SA that uses the P300's latency, amplitude scale, and reverse factor to transform signals. For evaluation, four datasets were introduced (two from conventional P300 Speller BCIs, one from a P300 Speller with face stimuli, and the last from a standard auditory oddball paradigm). *Results.* Although the standard approach without SA had an average precision (AP) score of 25.5%, the approach demonstrated a 35.8% AP score, and we observed that the number of subjects showing improvement was 36.0% on average. Particularly, we confirmed that the Speller dataset with face stimuli was more comparable with other datasets. *Significance.* We proposed a simple and intuitive way to align ERP signals that uses the characteristics of ERP signals. The results demonstrated the feasibility of cross-dataset transfer learning even between datasets with different paradigms.

1. Introduction

In the brain-computer interface (BCI) field, increasing models' classification performance and shortening the training time are important factors for building better BCI systems. Various advances have been made in electroencephalogram (EEG)-based BCIs during the past decade [1–4], but these issues remain challenging due to EEG's low signal quality and instability [5]. Various approaches can be used to overcome these problems. Trial-wise averaging is a primary strategy used to improve the signal-to-noise ratio (SNR). Having better stimulus/feedback types [6, 7] to modulate a clearer signal and using advanced signal processing methods [8, 9] for a more informative feature or better classification are also good approaches to attempt. However, several factors, including physiological and anatomical differences between people, environmental differences, and hardware characteristics, influence EEG [10].

BCI datasets have been shared recently [11, 12], and these open datasets have been used widely in BCI studies. Having more data is useful for constructing a well-trained model for good performance. In addition, these open datasets are useful for solving two important issues: improving performance and shortening calibration times.

Indeed, a recent P300 BCI speller study demonstrated that a deep learning model and a sufficiently large dataset produced good classification performance for unseen subject data [13]. With a cross-subject transfer learning approach, the authors used a convolutional neural network-based model referred to as 'EEGNet' [14]. A model trained with event-related potential (ERP) trials from 54 subjects was tested based on the remaining subjects' EEG. The performance was reported to be 89% (offline) and 85% (online). This was comparable to the conventional approach, which involved training and testing with the same subject's data. Inspired by this previous study [13], where subjects were classified based on the content learned within the same dataset (recorded under the same experimental paradigm), we studied to explore whether subjects in different datasets could also be classified using transfer learning.

Transfer learning is a technique that uses models trained with data from other domains and is one of the most critical techniques for solving the calibration time reduction issue [15]. The study [13] mentioned above is a good example of the way transfer learning is used in the BCI system. Fortunately, recent advanced models (EEGNet [14] and EEGInception [16]) appear to work with the transfer-learning concept for subject-to-subject transfer, although the way they work requires more in-depth investigation [17]. Other studies also attempted to decode neuropsychological parts with deep learning models [18–21]. As these studies showed, deep learning is emerging as a more robust method to use in BCIs.

However, in addition to subject-to-subject transfer, other factors must be considered. Normally, differences exist across datasets, as they are likely to be recorded with different devices (reference, electrode type, impedance, etc), paradigm parameters (inter-stimulus interval, stimulus type, feedback modality, etc), or experimental environments (sound/noise volume, illumination in the recording location, the refresh rate of the monitor, etc) [10, 15, 22, 23]. Due to these variations in datasets, signal characteristics can differ across datasets, and consequently, reusing the open datasets can be limited.

An ideal approach might involve attempting to identify an absolute latent space of signals from various datasets, as demonstrated in benchmark studies (e.g. word embedding [24] or autoencoder [25]). However, finding such a latent space and confirming its interpretability from a neuropsychological standpoint can be challenging. In contrast to other BCI paradigms, (e.g. motor imagery and steady state-visual evoked potential), P300-based BCIs distinguish target from non-target trials by using amplitude features that is well visible in the time domain. In the preprocessing phase, signal alignment (SA) is possible, and one can use the standard transfer-learning approach to build a model in the time domain. The features' variation is also likely presented in the time domain, which can be described as amplitude and latency, as reported in a P300 study [26]. Unfortunately, our current understanding is limited regarding the variability of ERP across different datasets. A more thorough comprehension of these ERP characteristics would be beneficial for data reuse and, subsequently, for constructing models using transfer-learning techniques across various datasets.

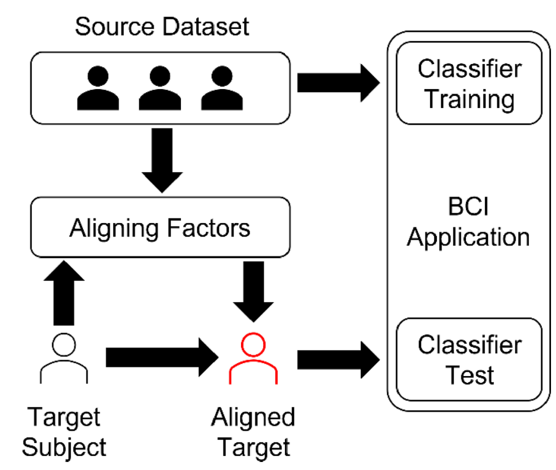


Figure 1. The conceptual procedure of signal alignment for transfer learning. The model, trained on the source dataset, is designed to classify the target subject by utilizing an aligning factor to transform the signals of the target subject into the same domain as the source dataset, thereby enabling better classification.

The principal idea of cross-dataset transfer is based on the domain generalization technique [27]. It is part of the deep learning method that is designed to create a robust model in two domains with different distributions [28]. The domains can be categorized into the source and target domains. When two domains are different, a model directly trained on the source domain does not show good performance for the target domain data. Similarly, classifying the EEG datasets without transferring (or adapting) raises the domain shift problem and is likely to fail because various EEG datasets are located in different domains. The best way to adapt or generalize the domains is to find a new latent space. However, constructing a latent space that is neurophysiologically well-understandable, is a challenging problem.

Thus, we attempted to adapt the target domain to the source domain using the ERP characteristics at the signal level, particularly P300 EEG signals. This process is referred to as SA through this study illustrated in figure 1.

In this study, we aimed to investigate different datasets' ERP patterns and propose a simple and intuitive SA method between different ERP datasets that can be used for preprocessing ahead of feature selection or classification. Additionally, we demonstrated the feasibility of the proposed SA method in dataset-to-dataset transfer learning.

The paper is organized as follows. In section 2, a new intuitive method, referred to as the SA method, is explained, and several datasets are introduced. In section 3, the results of our method are demonstrated. Then, we discuss our method's usefulness and certain findings about stimuli differences in the ERP paradigm in section 4. Finally, we conclude our study.

2. Methods and materials

2.1. SA

The P300's significant features are its amplitude and latency [26]. The ERP signal's latency varies from time to time, even within the same subject. Similarly, the signal's amplitude varies from person to person, and hardware characteristics, such as the amplifier gain or the dry/wet sensor types, affect it primarily [29]. In addition, the difference between the ground and the reference sensor locations can cause the signal to be reversed. Therefore, we selected these three significant SA factors (latency, amplitude scale, and reverse) to align the target data within the source domain.

First, the latency factor can be calculated using two ERP signals' cross-correlation. The cross-correlation calculates the similarity of two time-series data points given source data x and target data y, using the sliding dot product:

$$\hat{R}_{xy}(m) = \begin{cases} \sum_{n=0}^{N-m-1} x_{n+m} y_n^*, & m \geqslant 0 \\ \hat{R}_{yx}^*(-m), & m < 0 \end{cases} \tag{1}$$

$$R_{xy}(m) = \frac{1}{\sqrt{\hat{R}_{xx}(0)\,\hat{R}_{yy}(0)}} \hat{R}_{xy}(m) \tag{2}$$

where x_n is the signal at time point n, y^* denotes the complex conjugate of signal y, N is the length of the signals, m is the current sliding index with a range of $[-N+1, N-1]$, and $\hat{R}_{xy}(m)$ denotes the cross-correlation value at sliding index m. This should be normalized to prevent the different signals' amplitudes from affecting it. As shown in equation (2), the cross-correlation values are normalized so that the autocorrelations at the zero sliding index equal 1.

The $R_{xy}(m)$ is positively large if $x \approx y$, and it is negatively large if $x \approx (-y)$. Therefore, absolute value $|R_{xy}(m)|$ is larger if the signal x_{n+m} and y_n are similar. The optimal latency is calculated as follows:

$$\beta_{xy} = \underset{m}{\operatorname{argmax}} \left(|R_{xy}(m)|\right) \tag{3}$$

where β_{xy} denotes the optimal latency factor between signals x and y. The solution can be found using the $x$corr() function in the MATLAB software.

Second, the amplitude scale factor can be calculated as the ratio of the two signals' standard deviations. The standard deviation indicates how far the value deviates from its mean. This value can be considered an estimate of the signal power. Thus, the ratio of two standard deviations represents the ratio of amplitude differences, and it can be obtained with the following equation:

$$\alpha_{xy} = \frac{\sigma_x}{\sigma_y} \tag{4}$$

where σ_x is the standard deviation of signal x, and α_{xy} denotes the amplitude scale factor. In SA, this factor is multiplied to control the target signal's amplitude scale according to the source signal's scale.

Third, the reverse factor is determined based on the binary value indicating whether the signal should be inverted. This can be calculated by examining the sign of the cross-correlation value at the optimal latency using equation (2). If the cross-correlation at the optimal latency is negative, the reverse factor is negative, and positive otherwise. It is expressed as in equation (5) below:

$$\delta_{xy} = \begin{cases} 1, & R_{xy}(\beta_{xy}) \geqslant 0 \\ -1, & R_{xy}(\beta_{xy}) < 0 \end{cases} \tag{5}$$

where δ_{xy} denotes the reverse factor.

Finally, signal y can be aligned with x with three SA factors α_{xy}, β_{xy}, and δ_{xy}:

$$y_{\text{align}} = \delta_{xy} \odot \alpha_{xy} \odot y(t - \beta_{xy}) \tag{6}$$

where $\odot$ denotes the element-wise multiplication, and t is the time index. As a result, the domain of aligned signal y_{align} becomes similar to the domain of signal x.

2.2. SA for multi-dimensional data

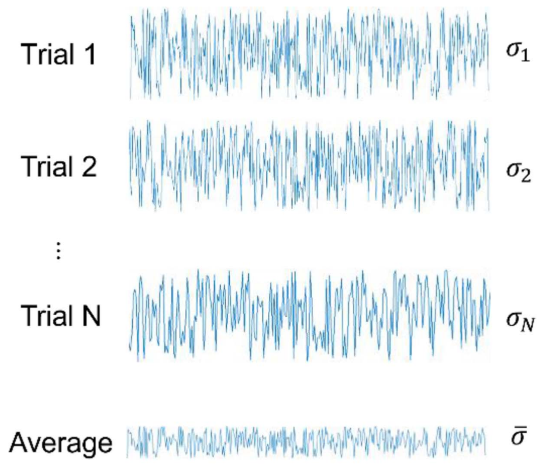
In general, EEG is multi-dimensional data: epochs, channels, and time points with many subjects. Thus, the method proposed should be extended to multi-dimensional data. Therefore, we calculated the latency, amplitude scale, and reverse factors somewhat differently.

Given index notations e (epochs), c (channels), t (time), and s (subject), we used $X_s \in \mathbb{R}^{e\times c\times t}$ and $Y \in \mathbb{R}^{e\times c\times t}$ rather than source signal x and target signal y. Representative template signal is required to determine the SA factors that align one dataset with another. This template can be obtained by averaging signals over epochs and subjects as in equation (7) for the source template and (8) for the target template,

$$X_{\text{template}} = \frac{1}{S}\sum_{s=1}^{S}\left(\frac{1}{E}\sum_{e=1}^{E} X_s\right) \in \mathbb{R}^{c\times t} \tag{7}$$

$$Y_{\text{template}} = \frac{1}{E}\sum_{e=1}^{E} Y \in \mathbb{R}^{c\times t} \tag{8}$$

where S denotes the number of all subjects in the source dataset, and E denotes the number of epochs for each subject. We applied equation (3) to calculate the latency with this template signal and obtain



$$\bar{\sigma} < \frac{1}{N}\sum_{i}^{N}\sigma_i$$

Figure 2. Graphical illustration of the effect of averaging to signal variance. As shown, the averaging dcreases the signal power, consequently resulting in a smaller standard deviation ($\bar{\boldsymbol{\sigma}}$) compared with the standard deviation ($\boldsymbol{\sigma}_i$) of each trial.

latency vector $\left[\beta_{xy}^1, \beta_{xy}^2, \ldots, \beta_{xy}^c\right]^{\mathsf{T}}$. T denotes the transpose of the vector. Assuming that the dataset-dependent jitter affects all of the channels equally, the final optimal latency factor, β_{xy}, is calculated as follows:

$$\beta_{xy} = \frac{1}{C}\sum_{c=1}^{C}\beta_{xy}^c \tag{9}$$

where C denotes the number of channels.

Individual epochs rather than the template signal was used for amplitude scale factor vector $\boldsymbol{\alpha}_{xy} = \left[\alpha_{xy}^1, \alpha_{xy}^2, \ldots, \alpha_{xy}^c\right]^{\mathsf{T}}$. The amplitude scale factor is calculated using non-target signals because the goal is to get the default amplifier gain. As averaging many non-target signals results in a zero-like waveform with a smaller amplitude scale, the amplitude scale factor may be close to zero as expected as illustrated in figure 2. Thus, it is better to use the signals' epoch-wise standard deviations ($\boldsymbol{\sigma}_{x,s,e}$ and $\boldsymbol{\sigma}_{y,e}$). Amplitude scale vector $\boldsymbol{\sigma}_x \in \mathbb{R}^{c\times 1}$ and $\boldsymbol{\sigma}_y \in \mathbb{R}^{c\times 1}$ can be obtained by averaging the epoch-wise standard deviations over epochs as well as subjects based on the following equations:

$$\boldsymbol{\sigma}_x = \frac{1}{S}\sum_{s=1}^{S}\left(\frac{1}{E}\sum_{e=1}^{E}\boldsymbol{\sigma}_{x,s,e}\right), \ \boldsymbol{\sigma}_y = \frac{1}{E}\sum_{e=1}^{E}\boldsymbol{\sigma}_{y,e} \tag{10}$$

Finally, through applying equation (4), the element-wise ratio can be obtained for amplitude scale factor vector $\boldsymbol{\alpha}_{xy} = \boldsymbol{\sigma}_x/\boldsymbol{\sigma}_y$.

The reverse factor is calculated based on the latency vector because it is intended to identify a negative relation per channel. Therefore, the reverse factor vector is $\boldsymbol{\delta}_{xy} = \left[\delta_{xy}^1, \delta_{xy}^2, \ldots, \delta_{xy}^c\right]$ and is multiplied by the amplitude scale vector.

The entire pipeline used to obtain the aligned target signal is described in algorithm 1.

2.3. Datasets and preprocessing

We employed four EEG datasets to evaluate the method proposed. Three were recorded with the P300 speller systems [30–32], and the last was obtained with the auditory oddball paradigm [11]. Detailed information on each dataset is provided in table 1.

The first dataset [30] is derived from the BCI Competition III Dataset II, which we refer to as the 'Speller-1' dataset in this study. The EEG was recorded from two subjects using the standard row/column-based BCI speller system. The speller paradigm was designed and conducted with BCI2000 software, and signals were acquired via 64 EEG electrodes at a 240 Hz sampling frequency. Ear-referencing was used, and bandpass-filtering at 0.1 Hz to 60 Hz was performed. Each subject performed two training and three test sessions.

The second dataset [31], referred to as 'Speller-2,' was also acquired with the standard P300 BCI speller system using BCI2000 software with ActiveTwo (Biosemi Inc.). The EEG data were recorded from 55 subjects using 64 electrodes at 512 Hz, and signals were processed using common average referencing (CAR) and bandpass-filtering (0.1 Hz to 60 Hz).

The third dataset [32] (Speller-face) was also recorded using the P300 BCI system but with certain differences in the experimental paradigm. The paradigm of the Speller-1 and Speller-2 datasets designed the symbols (A to Z, 1–9, and _) with six rows and six columns, and each row and column blinked. However, in the Speller-face dataset's paradigm, rows and columns blinked irrespectively and randomly [6], and the subject was also shown a face image rather than the symbols when the visual stimulus blinked [33]. The EEG data were recorded using BrainAmp (Brain Product, Inc.), which was composed of 62 electrodes at a rate of 1000 Hz, nasion-referenced, and grounded to electrode AFz.

The last dataset [11], referred to as 'Aud-Oddball,' is the auditory oddball paradigm, which is 70% the standard stimulus (500 Hz tone), 15% the oddball stimulus (1000 Hz tone), and 15% the distracter stimulus (1000 Hz tone with white noise). The distracter stimulus was not used in this study due to its potential to make irrelevant signals [34]. EEG was originally recorded using 64 channels at 1024 Hz and

Algorithm 1. Signal alignment method.

Input: source dataset X_s, $s \in \{1,\dots,S\}$ and target dataset Y
Output: aligned target subject Y_{align}
1: Calculate X_{template} and Y_{template} (equations (7) and (8))
2: Get the optimal latency β_{xy} using X_{template} and Y_{template} (equation (9))
3: Calculate $\boldsymbol{\sigma}_x$ and $\boldsymbol{\sigma}_y$ using X_s and Y (equation (10))
4: Get the amplitude scale $\boldsymbol{\alpha}_{xy}$ (equation (4))
5: Get the reverse factor $\boldsymbol{\delta}_{xy}$ using the optimal latency vector (equation (5))
6: Calculate the Y_{align} using $\boldsymbol{\alpha}_{xy}$, β_{xy}, $\boldsymbol{\delta}_{xy}$ (equation (6))

Table 1. The detailed information of four ERP datasets. (*All subjects have different numbers of trials. In particular, one subject has 1350 trials, and the average number of trials over other subjects is 1900.).

Dataset	Stimulus type	Acquisition device	Sampling rate	Number of channels	Number of subjects	Number of training trials per subject	Number of test trials per subject	Target-non-target ratio
Speller-1 [30]	Row/Column blinking	—	240 Hz	64	2	15 300	18 000	1:5
Speller-2 [31]		Biosemi ActiveTwo	512 Hz	64	55	1800	5040	1:5
Speller-face [32]	Random-set presenatation [6], Face blinking [33]	Brain products brainAmp	1000 Hz	62	54	3960	4320	1:5
Aud-Oddball [11]	Sound (500 Hz/1000 Hz/1000 Hz + noise)	Biosemi ActiveTwo	1024 Hz	64	13	1350/1850–1950*		1:4.6

downsampled at 256 Hz to reduce the data size. We introduced this dataset to compare transfer learning possibilities between stimulus paradigms.

All data were re-referenced using CAR, and 20 common channels (Fz, T7, C3, Cz, C4, T8, CP5, CP1, CP2, CP6, P7, P3, Pz, P4, P8, PO3, PO4, O1, Oz, O2) were selected to reduce the computing time and equalize the channels. Then, the signals were bandpass-filtered in the cutoff frequency range of 0.5 Hz to 30 Hz and epoched from the beginning of the stimulus to 800 ms. Thereafter, the signals were resampled to 40 Hz to equalize the sampling frequency and reduce computation cost. This change has the same effect as filtering the signal up to 20 Hz with a lowpass filter, based on the Nyquist theorem, and the P300 signal is not affected because it consists of lower frequencies.

The signals were averaged in random combinations to determine whether the performance improved as the SNR increased. We created five cases and averaged a random combination of 1, 5, 10, 15, and 20 epochs before the signals were resampled. As the total number of epochs remained unchanged, certain epochs were duplicated in the averaged signals, except for in the case of 1—a single-trial classification.

2.4. Evaluation

Although a model may appear to be very suitable for solving a specific problem, sometimes it cannot overcome the problem with other datasets. To avoid this situation, we used all combinations of the four datasets. We had 12 cases of transfer learning. Each dataset was used to train the classifier, and other datasets were used to test the method suggested. In addition, each case was tested by averaging the trials as mentioned above. In summary, 60 results were simulated.

Before the model attempted to classify the data, we checked the method proposed quantitatively. Because the SA method adapts datasets in the time domain, time-related quantitative measurements are needed, and correlation is one of the ways to obtain these quantitative measurements. The correlation is calculated with the source template signal and target or aligned target signal. As expected, the aligned target's correlation was greater than that with the target signal due to the proposed method's alignment.

2.5. Metrics

Appropriate metrics should be selected to evaluate the method proposed with the classifier. Because the datasets have imbalanced target and non-target ratios, accuracy cannot explain the result accurately. Many studies match the number of targets and non-targets by discarding the non-target trials or using other metrics to evaluate the imbalanced situation [35, 36]. The primary approach is to use a confusion matrix that shows true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN). Various measures construct this confusion matrix. The formula for accuracy is $(\mathrm{TP}+\mathrm{TN})/(\mathrm{TP}+\mathrm{TN}+\mathrm{FP}+\mathrm{FN})$. The imbalanced data result in greater accuracy because the true negative carries considerable weight. Precision and recall are used to avoid the true negative's influence. Precision is calculated with $\mathrm{TP}/(\mathrm{TP}+\mathrm{FP})$, and recall is calculated with $\mathrm{TP}/(\mathrm{TP}+\mathrm{FN})$. Using these measures, the f1-score is calculated as the harmonic mean of precision and recall. Because they do not consider the number of true negatives, the f1-score can be a good metric for imbalanced data.

Another issue with the metric is the threshold. Most test models yield probability values and usually classify them as true if the probability is greater than 0.5, and false otherwise. However, sometimes it is helpful to have different thresholds, and the receiver operating characteristic (ROC) curve is the usual method for applying this. The ROC curve calculates the recall and false-positive rate ($\mathrm{FPR}=\mathrm{FP}/(\mathrm{TN}+\mathrm{FP})$) by changing the threshold from 0 to 1. The area under the ROC curve (AUC-ROC) is used to quantify the ROC curve and provides a model-wise evaluation of binary classifiers. However, the ROC does not work well with imbalanced data because it is also affected by the true negative. To avoid its influence, the curve graph is drawn with precision and recall, which is referred to as a precision-recall curve (PRC), and the area under the PRC curve (AUC-PRC) is used as a metric. It is also referred to as average precision (AP), calculated as follows:

$$\mathrm{AP}=\sum_{n}(R_n-R_{n-1})P_n \tag{11}$$

where P_n and R_n are the precision and recall at the nth threshold. This shows the model's ability to classify the true positive accurately, and it contains more information than the f1-score [37].

2.6. Data quality and statistical test

The data were classified using the step-wise linear discriminant analysis (SWLDA) model to test the method proposed [38]. The model was trained and tested within each subject to assess the datasets' baseline performance. Thereafter, the model was trained with all subjects in a dataset, and we tested a subject in another dataset with and without applying the SA method, respectively.

Speller datasets include training and test sessions, whereas the Aud-Oddball dataset does not. Therefore, the epochs of this dataset were divided into training and test data with a ratio of 7:3 because no distinction existed between the training and test. Typically, no difference exists between the training and test sessions in any of the datasets, but the Speller-1 dataset has certain differences. Figure 3 demonstrates the averaged target and non-target signals over all subjects in the training and test sessions. As the figure shows, the qualities of datasets differ. To assess how well the SWLDA model can classify the data, the model trained with the training data from one subject tested that subject's test data. Table 2 shows the AP scores and standard deviation of each dataset obtained from the SWLDA model. As the ERP pattern was not obtained from the test data, the Speller-1 dataset's training data were divided into training and test data at a ratio of 7:3, like the Aud-Oddball dataset. This dataset was used as Speller-1 in this study, and the original test dataset was referred to as Speller-1-all. The larger the number of averaging trials, the greater the SNR, and the better the signal quality; thus, the SWLDA model classified better.

The chance level is demonstrated in table 2, but some studies [39, 40] reported that statistical chance level varies depending on the sample size and suggest using a sufficient number of samples. In this situation, training and testing in a dataset, each dataset had 1850–5040 testing trials (except for the Speller-1 dataset) with 10–55 subjects. Hence, the sample size was sufficient to compare with the theoretical chance level. Thus, we simply used the theoretical chance of AP, which is the ratio between the target and non-target sample sizes for comparison with the performance analysis. Additionally, for comparing performances among methods, we used a paired t-test with a significance level of 0.05.

2.7. Comparison with other methods

In the related study [41], covariance matrix centroid alignment (CA) was introduced for SA, which typically uses the covariance matrices and the mean of them. Also, various types of means can be used, such as the Euclidean mean for Euclidean alignment [42], the Riemannian mean for Riemannian alignment [43], and the log-Euclidean mean for log-Euclidean alignment [41]. In this study, we used CA with the log-Euclidean alignment used for the comparison study. For all possible cases with 20 epochs averaged, we computed the AP scores from the CA method and compared them with the performance of SA.

3. Results

3.1. SA

Figure 4 shows the representative cases after the signals were aligned using the SA method. In figure 4(a), the successful case of SA is presented, which showed the greatest increase in AP from Speller-2 to Speller-face dataset transfer learning. The correlation coefficients between the source and non-aligned target or

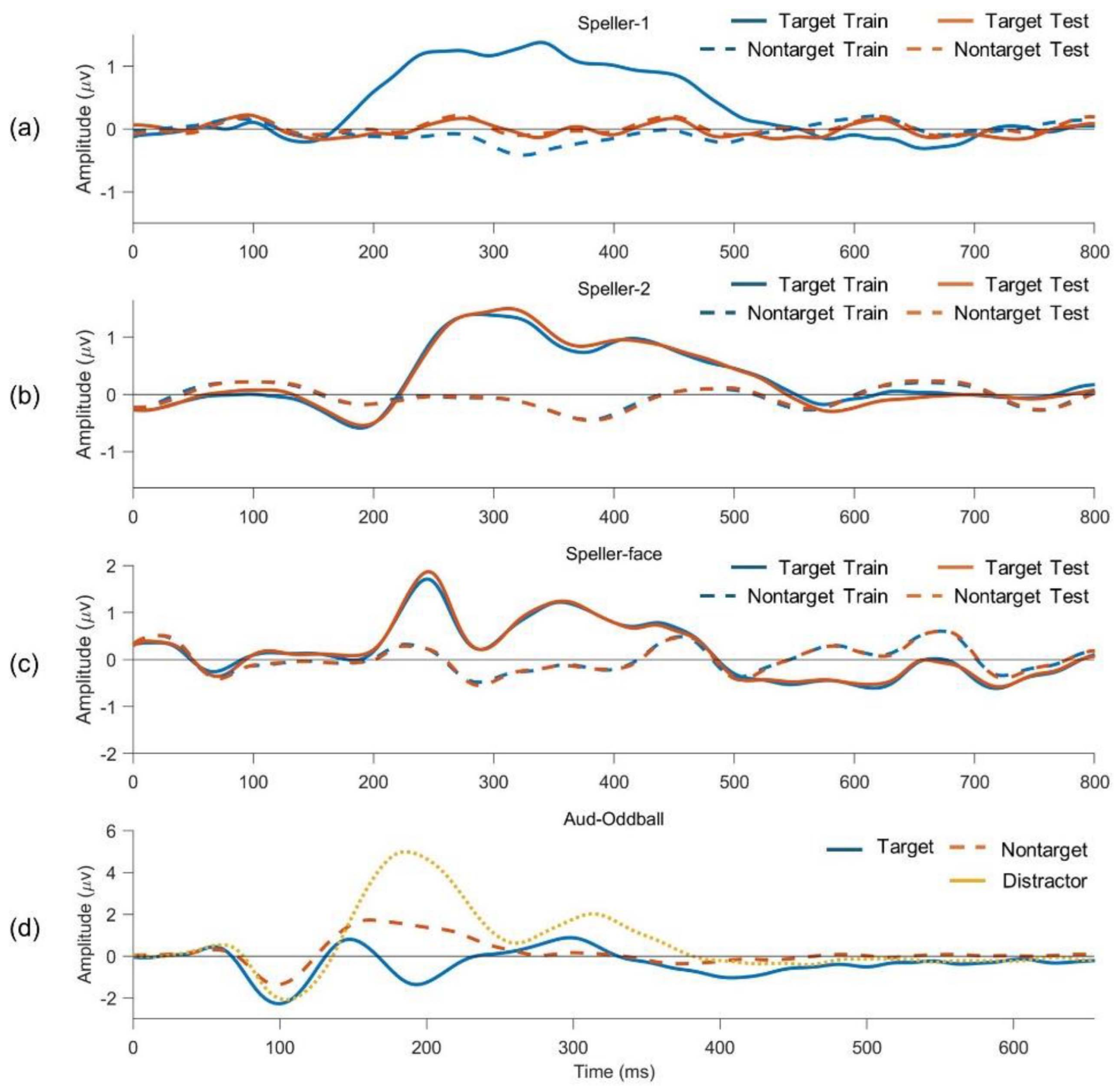


Figure 3. Grand averaged ERP signals at Cz. (a) Speller-1, (b) Speller-2, (c) Speller-face, and (d) Aud-Oddball. Solid lines denote target data, whereas dashed lines are non-target data. Particularly, the distractor is represented by a dotted line in (d).

Table 2. Average precision scores within the dataset. The values presented represent the mean and standard deviation. Significant results are observed in all cases except for Speller-1-all.

The number of epochs averaged	1 epoch	5 epochs	10 epochs	15 epochs	20 epochs	Chance Level
Speller-1-all	17.4% ± 0.4%	17.9% ± 0.1%	18.7% ± 1.0%	19.6% ± 0.4%	19.2% ± 1.0%	16.7%
Speller-1 (only the train dataset was used)	40.6% ± 8.6%	72.6% ± 17.0%	87.8% ± 9.4%	92.3% ± 8.4%	89.4% ± 14.6%	18.4%
Speller-2	51.9% ± 11.9%	78.2% ± 16.5%	90.2% ± 13.7%	94.3% ± 11.0%	97.3% ± 6.8%	16.8%
Speller-face	83.8% ± 12.6%	97.9% ± 4.6%	99.2% ± 3.4%	99.7% ± 2.0%	99.6% ± 3.2%	16.8%
Aud-Oddball	72.0% ± 19.9%	85.0% ± 18.1%	95.4% ± 9.8%	93.4% ± 12.9%	98.9% ± 2.1%	18.4%

aligned target signals were $r = 0.10$ and $r = 0.55$, respectively. As seen, the latency of the P300 peak of the target signal is aligned with the source signal's peak, and the amplitude is also adjusted close to the source signal. The target signals at Fz and Pz were reversed, and all signals were shifted approximately 50 ms.

In contrast, figure 4(b) shows an unsuccessful case of SA from the Speller-face to Speller-2 datasets (source vs. target: $r = 0.37$, source vs. aligned: $r = 0.26$). The Fz channel's target signal is aligned with the N400 components and unexpectedly reversed. The Pz channel's target signal moved regardless of the source signal's P300 because it did

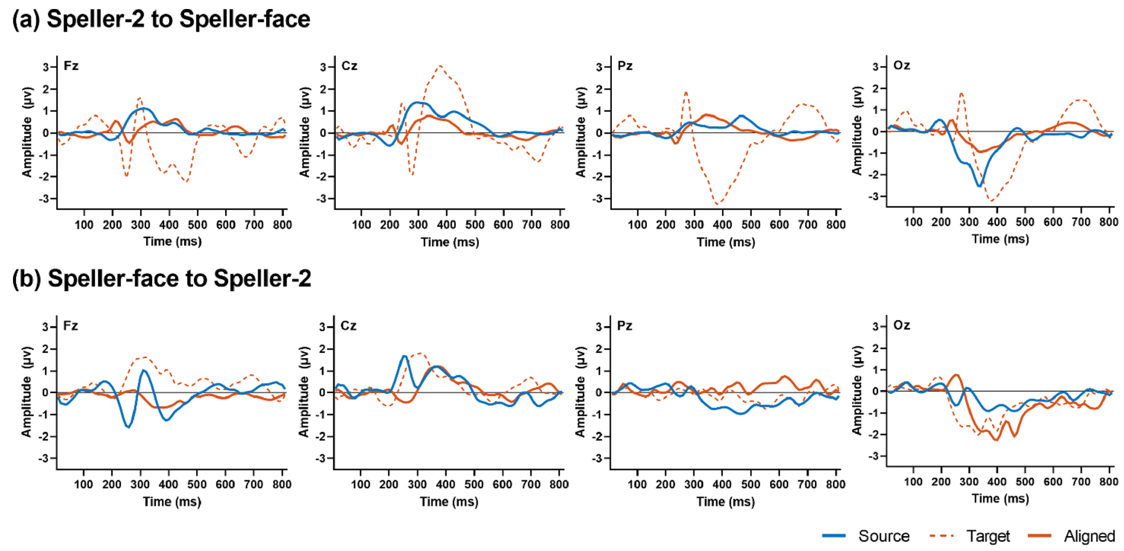


Figure 4. Representative ERP signals of successful and unsuccessful cases from signal alignment using SA. (a) A successful case in 'Speller-2 (source) to Speller-face (target).' (b) An unsuccessful case in 'Speller-face (source) to Speller-2 (target).' Each figure includes the grand averaged signals of the source (blue line), target (red dotted line), and aligned target (red line).

Table 3. Correlation coefficient values of source and target signals without alignment (standard) and with SA. The results are obtained from ERP of 20 epochs average, and only subjects who show $p < 0.05$ in SA are included. Bold indicates cases of improved correlations in SA.

Test (to) / Train (from)	Speller-1		Speller-2		Speller-face		Aud-Oddball	
	Standard	SA	Standard	SA	Standard	SA	Standard	SA
Speller-1			0.41 ± 0.32	-0.07 ± 0.51	$\mathbf{-0.08 \pm 0.41}$	$\mathbf{0.22 \pm 0.41}$	0.02 ± 0.46	-0.15 ± 0.47
Speller-2	0.39 ± 0.32	-0.02 ± 0.62			$\mathbf{0.03 \pm 0.40}$	$\mathbf{0.25 \pm 0.42}$	0.00 ± 0.42	-0.09 ± 0.48
Speller-face	$\mathbf{-0.04 \pm 0.48}$	$\mathbf{0.35 \pm 0.38}$	$\mathbf{0.02 \pm 0.43}$	$\mathbf{0.22 \pm 0.40}$			$\mathbf{-0.21 \pm 0.42}$	$\mathbf{0.34 \pm 0.38}$
Aud-Oddball	-0.06 ± 0.45	-0.06 ± 0.55	0.01 ± 0.41	-0.09 ± 0.46	-0.04 ± 0.41	0.08 ± 0.46		

not show the P300 signal clearly. Certain differences exist in amplitude and latency. The amplitude does not have any significant abnormality because it is calculated using the signal's standard deviation. As seen, the signal quality appears to be poor when P300 is not visible, such as the signal at Pz in figure 4(b).

Table 3 demonstrates the correlation coefficients between the source and target signals of standard or SA. In this calculation, ERP averaged over 20 epochs, and only subjects who showed significant correlation coefficients ($p < 0.05$) in SA were included. As seen, some cases showed a meaningful improvement in SA. Particularly, the dataset 'Speller-face' appeared to be highly compatible with the other datasets. The correlation coefficients in all datasets improved from Speller-face to other datasets: $r = 0.35$ (to Speller-1), $r = 0.22$ (to Speller-2), and $r = 0.34$ (to Aud-Oddball). Furthermore, the opposite direction (to Speller-face) demonstrates improvement: $r = 0.22$ (from Speller-1) and $r = 0.25$ (from Speller-2). The channel-by-channel results can be found in appendix and show a more detailed trend.

3.2. Performance in transfer learning

The results of evaluating cross-dataset transfer learning are presented in table 4. Transferring from other datasets to Speller-1 showed performance improvement in some cases, for example, 44.7% (from Speller 2, epochs = 20), 20.2% (from Speller-face, epochs = 20), and 22.5% (from Aud-Oddball, epochs = 20). However, due to the small number of subjects ($N = 2$), statistical significance could not be tested.

Overall, performance improved in most cases except for in the cases of 'Speller-face to Speller-2' and 'Speller-2 to Aud-Oddball.' Particularly, transferring to Speller-face from all datasets showed the most meaningful improvement in performance. For 20 epochs, the averaged, standard, and SA methods showed APs of 18% to 36.5% (from Speller-1), 17.4% to 35.5% (from Speller-2), and 14.3% to 34.6% (from Aud-Oddball). This is consistent with the results of alignment quality in the previous section.

Based on the collection of all of the performance percentages for 20 averaged epochs, the average AP and standard deviation were 25.5% ± 17.6% for standard and 35.8% ± 24.6% for SA, and statistical analysis revealed that the difference was significant ($p < 0.05$).

Figure 5 presents the changes in every subject in each case. As expected, a large population that

Table 4. The average precision scores in transfer learning. Transfer learning from other datasets to Speller-face demonstrates promising results; however, the reverse transfer learning from Speller-face to other datasets does not yield the same success. This suggests that Speller-face might have learned factors beyond P300 in its training process. The results were tested by the paired t-test.

	Test (to)	Speller-1-train		Speller-2		Speller-face		Aud-Oddball	
Train (from)	Epochs averaged	Standard (%)	SA (%)	Standard (%)	SA (%)	Standard (%)	SA (%)	Standard (%)	SA (%)
Speller-1	1			23.7	23.3	**16.3**	**20.7** [a]	20.0	23.9
	5			35.8	33.5	**18.2**	**24.2** [a]	23.0	30.1
	10			**29.9**	**37.0** [a]	**14.5**	**35.4** [a]	22.8	38.3
	15			52.0	46.3	**21.3**	**41.6** [a]	28.8	46.6
	20			41.9	46.8	**18.0**	**36.5** [a]	**28.7**	**60.4** [a]
Speller-2	1	20.4	19.8			**16.1**	**19.6** [a]	20.5	20.1
	5	23.8	25.2			**17.1**	**24.6** [a]	27.4	27.7
	10	26.8	29.8			**19.2**	**32.4** [a]	29.3	34.5
	15	35.6	34.0			**17.8**	**33.9** [a]	27.5	33.2
	20	31.2	44.7			**17.4**	**35.5** [a]	35.7	45.5
Speller-face	1	17.3	17.7	19.0	17.9 [a]			**15.5**	**20.3** [a]
	5	17.5	16.8	24.8	19.6 [a]			**15.1**	**28.2** [a]
	10	16.8	16.7	28.1	22.2 [a]			12.9	26.1
	15	14.9	20.0	32.8	26.3 [a]			**13.8**	**27.0** [a]
	20	17.3	20.2	35.8	25.6 [a]			15.8	29.6
Aud-Oddball	1	16.9	17.9	**17.5**	**18.6** [a]	**15.3**	**20.1** [a]		
	5	15.6	19.0	21.7	21.9	**14.5**	**23.7** [a]		
	10	17.1	24.0	**19.3**	**23.3** [a]	17.0	22.5		
	15	16.5	19.5	24.4	26.6	**13.0**	**29.6** [a]		
	20	17.9	22.5	**24.2**	**29.6** [a]	**14.3**	**34.6** [a]		

[a] Statistically significant cases are marked with * ($p < 0.05$) and improvement in SA is marked with bold font.

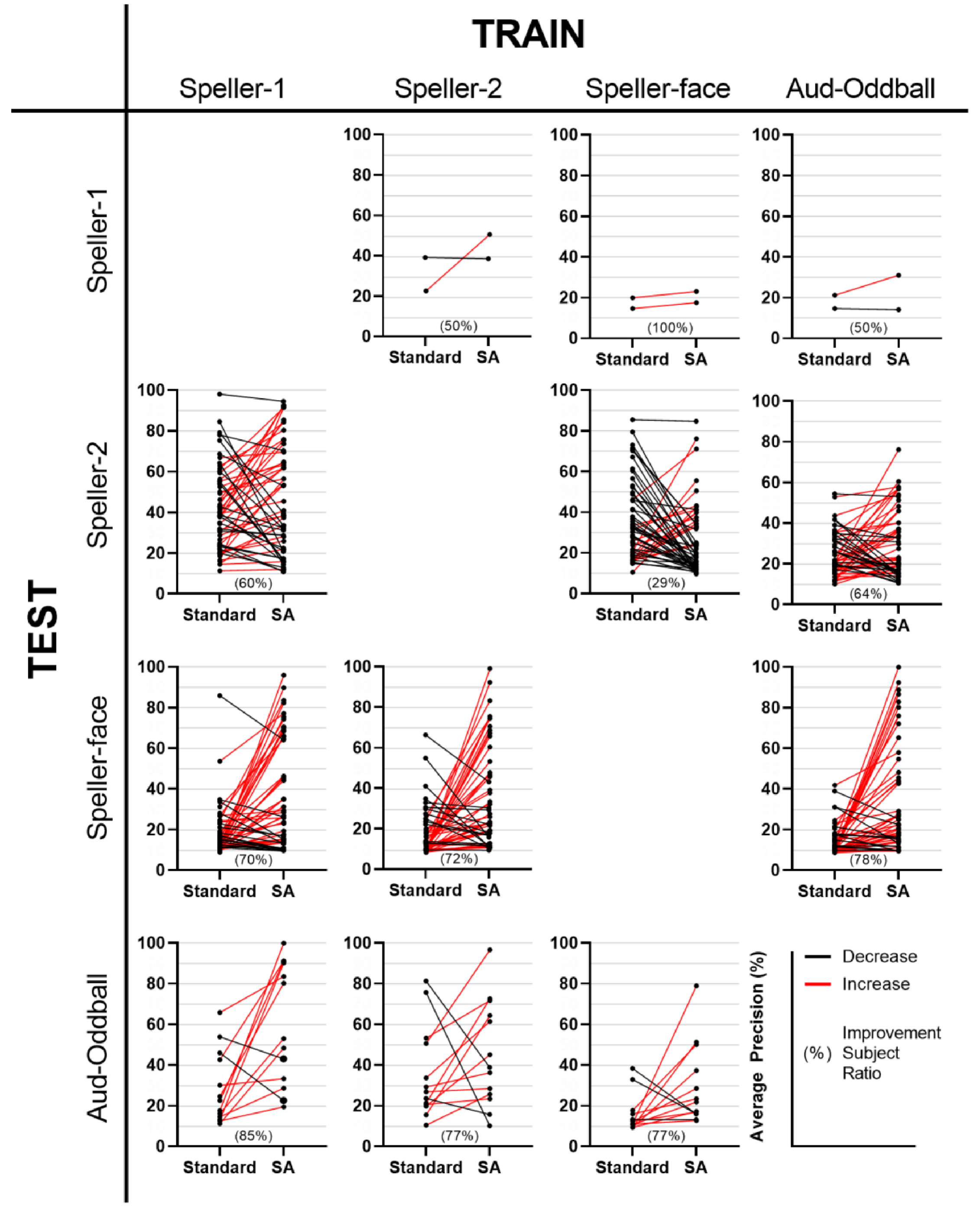


Figure 5. Average precision change and the number of subjects who improved. The figure illustrates alterations in average precision values before and after applying the SA method from 20 averaged epochs. Each dot represents a subject, with cases of improved performance highlighted by red lines. The ratio of subjects demonstrating enhanced performance is indicated within each figure.

demonstrated improved performance was observed in some cases, whereas others did not. The population rates with improved performance are marked in each figure. The improved performance range was from a minimum of 0.1% to a maximum of 89.7%, and the average was 25.0%.

3.3. Comparison with CA method

Table 5 shows the comparison results between CA and SA methods. The results tested via the Speller-face dataset show statistical differences; for example, CA and SA show 13% and 37% (from Speller-1), 20% and 36% (from Speller-2), and 13% and 35% (from Aud-Oddball). In addition, the result from Speller-1 to Aud-Oddball shows 27% with CA and 60% for SA. The other result from Speller-face to Speller-2 shows 13% for CA and 26% for SA. The CA method shows lower standard deviations than the SA method except for in some cases. These cases were confirmed significant using the paired t-test ($p < 0.05$).

Table 5. The average precision scores for CA and SA. The bold results with an asterisk * indicate statistically significant differences ($p < 0.05$, tested by the paired t-test). In these cases, the SA method yields better results than the CA method.

Train (from) \ Test (to)	Speller-1		Speller-2		Speller-face		Aud-Oddball	
	CA (%)	SA (%)	CA (%)	SA (%)	CA (%)	SA (%)	CA (%)	SA (%)
Speller-1			52 ± 20	47 ± 27	**13 ± 9**	**37 ± 26***	**27 ± 23**	**60 ± 30***
Speller-2	40 ± 1	45 ± 0			**20 ± 6**	**36 ± 25***	49 ± 33	46 ± 20
Speller-face	17 ± 18	20 ± 20	**13 ± 43**	**26 ± 17***			35 ± 15	30 ± 26
Aud-Oddball	17 ± 1	23 ± 10	32 ± 41	30 ± 16	**13 ± 11**	**35 ± 27***		

Table 6. The average precision scores of transfer learning averaged from 20 trials without the reverse factor. The Speller-1-train dataset was not tested because the results are the same as in table 4 due to the tiny population.

Train \ Test	Speller-2		Speller-face		Aud-Oddball	
	Standard (%)	SA (%)	Standard (%)	SA (%)	Standard (%)	SA (%)
Speller-1	**42.0**	**59.9** [a]	18.0	16.0	26.9	25.8
Speller-2			**17.1**	**35.1** [a]	34.5	22.8
Speller-face	36.0	27.6 [a]			14.7	17.7
Aud-Oddball	24.5	26.0	14.2	16.2		

[a] Statistically significant cases are marked with * ($p < 0.05$) and improvement in SA is marked with bold font.

4. Discussion

4.1. Analysis of the cross-dataset transfer learning

Some interesting points emerged in the results. In the case of Speller-1 to Speller-2 or Speller-2 to Speller-1, the correlation values changed from positive to negative in many channels (see appendices 1(a) and (b)), but the performance did not improve. Thus, the reverse factor may not work in these cases, as the reverse factor that the cross-correlation calculates appears to be inaccurate in some cases. To check about the reverse factor's effects, the SWLDA model computes the result without a reverse factor. This computation used the data average of 20 trials, and the random seed was changed. According to table 6, Speller-1 to Speller-2 and Speller-2 to Speller-face showed improvements in AP, but the other results differed. This shows that the reverse factor can interfere with the alignment and needs improvement. To ensure that the reverse factor is applied accurately, it is necessary to use a small number of trials in the calibration session to check whether the reversing improves the model.

The other interesting point is that the model trained using the Speller-1 or 2 datasets worked well on the Speller-face dataset with the SA method but not in the opposite direction. The speller datasets have the same paradigm, but the Speller-face dataset has certain variations, such as random-set presentation and face blinking [6, 33]. Due to the difference in stimulus type, the ERP pattern may differ. The ERP signals appeared to be the same when focused on a specific stimulus. Still, the Speller-face dataset could have additional components to recognize a face object. Figure 3 also shows that the signal form differed from that in the Speller-1 or 2 datasets, similar to the pre-negative peak, late component, etc, which appears when a person recognizes a face [44, 45]. Therefore, the SWLDA model, trained with only the P300 components in the Speller-1 and Speller-2 datasets, could capture the P300 in the Speller-face dataset and offer good results. Conversely, the model that learned additional information about face recognition in the Speller-face dataset produced poor results because it needed that information to classify the Speller-2 dataset. To interpret the features' importance, we investigated the features that the model selected as shown in figure 6. The Speller-1 dataset's features were selected primarily between 200 milliseconds (ms) and 500 ms, and the Speller-2 dataset's features were selected between 100 ms and 500 ms. The frontal and occipital channels were selected for both datasets, but the temporal, central, and frontal channels were selected between 100 ms and 400 ms for the Oddball dataset. Largely, the Speller-face dataset selected central channels between 0 ms and 500 ms.

ERP signals are susceptible to the type of stimulus. The processing method in the brain varies depending upon which path the stimulus derives from among the five senses, and even if the stimulus is the same, the signal may vary depending on personal differences, such as familiarity. Figure 6 implies that even if the same paradigm is used, ERP signals may vary if the stimuli differ, and transfer learning may be difficult. Conversely, the results of this study suggested that datasets with the same stimulus can be used for transfer learning, even if the paradigms differ.

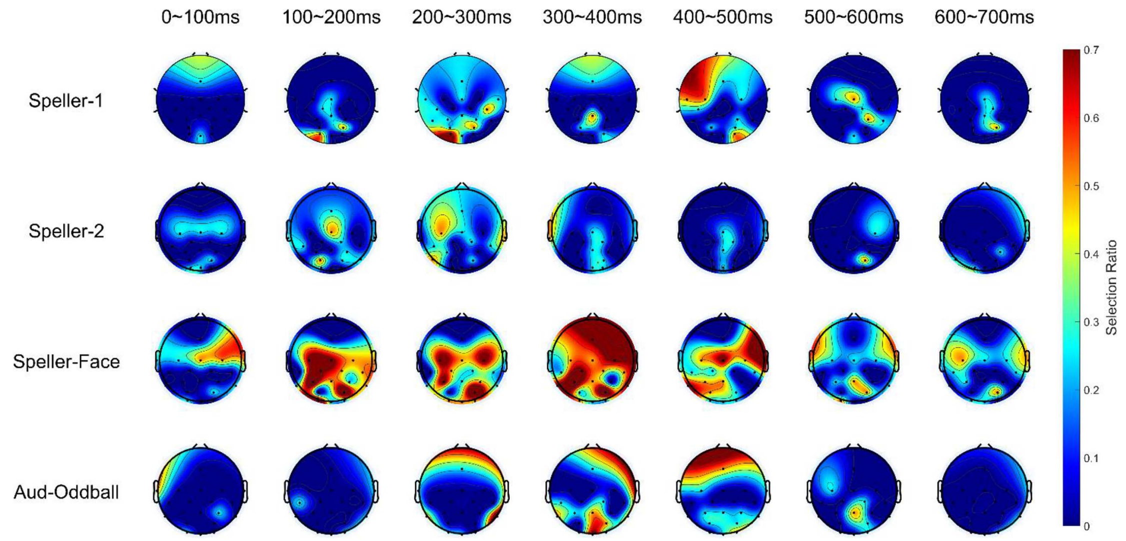


Figure 6. Features selected by classification model. The topoplot depicts the ratio of the average number of features selected over a 100 ms interval. For instance, a value of 0.5 between 100 ms and 200 ms on a channel indicates that the SWLDA model selected half of the features within that time range.

4.2. Comparison with CA method

The suboptimal performance of the CA method may stem from its reliance on the covariance matrix, which, by its nature, may not fully encapsulate the temporal intricacies inherent in the P300 signal. Given the significance of temporal dynamics in the P300 component, the failure to effectively incorporate this information could hinder the method's ability to discern relevant patterns for accurate classification. Consequently, the model's classification task is more challenging, as it struggles to discern meaningful distinctions within the data. Thus, the insufficient consideration of temporal features within the CA approach likely contributes to its diminished efficacy in discriminating P300-related activity.

4.3. Methodological improvements

One of the ways to improve the proposed SA method further is to add features other than the three factors (latency, amplitude scale, and reverse). One potential candidate is the 'stretch factor.' Each person has a different time when the P300 reaches its peak after the N200 and a different time for it to return to ready. Someone's P300 signal begins at 200 ms and ends at 400 ms, whereas another person's P300 signal begins at 250 ms and ends at 500 ms. In this case, it is not easy to match the signal by simply changing the latency, and transforming the signal may need a method that stretches the signal for perfect matching. As a solution, using phase information, such as the phase locking value or the dynamic time warping, can be a new alignment method. However, it is still complex to compute the signals epoch by epoch with these methods.

Another candidate is to specify 'the P300 range.' Some signals are aligned with N200 because the N200 is more potent than P300; thus, the target signal moved more than the number of points expected. To avoid this, it would be helpful to specify the P300 range to calculate a more accurate latency factor. As mentioned above, it can help determine how much to stretch the signal. However, the problem is that a new algorithm must be designed to specify the P300 range within a subject.

From the perspective of signal quality, the BCI speller paradigm has a very short inter-stimulus interval. It is important to set the epoching range or drop some epochs to prevent the target data, including the non-target ERP range [46]. The Speller-1 and 2 datasets' stimuli blink at 5.3 Hz, and the Speller-face dataset blinks at 7.4 Hz. Because the epoching range was set from 0 ms to 800 ms in this study, some non-target epochs include the target ERP information. This can prevent the model from selecting the pure features; thus, the result may be contaminated.

Another improvement point is related to classifier models. Although the SA method improves P300 classification performance, the highest AP score still needs to be higher than training and classifying within one dataset. This is attributable to the characteristics of the SWLDA model, which selects and compares values only at specific points in the signal. Thus, even if a certain amount changes the latency, jitter may affect it, and the classification may not be accurate. In some situations, ensuring the features that the SWLDA model selects is complex. In figure 6, the SWLDA model selects many features in the Speller-face dataset, whereas the other datasets select only a small number of features. Because the SWLDA model selects the features based on the p-values, the problem of statistical testing still applies to the model. As the sample size becomes extremely large, the confidence limit decreases, and even with a slight difference, the p-value converges to 0 [47]. The Speller-face dataset has 35 640 target signals and 178 200 non-target signals; hence, most p-values are computed as 0, and the SWLDA model selects most of the features. Therefore, the features selected are supersets, not optimal sets, and may not offer the best performance.

To enhance performance, alternative classifiers can be explored. However, given the large number of features, conventional machine learning techniques, such as LDA and SVM, may not be optimal choices, necessitating careful feature selection. Deep learning models, as demonstrated in prior studies [13, 48], offer promise in both feature selection and classification tasks. Thus, investigating the synergies between SA and classification methods (xDAWN [9], Riemannian geometry [49], and deep learning models) is an interesting topic. Nonetheless, considering the focus of this study is on introducing a straightforward SA method and demonstrating its feasibility in cross-dataset transfer, comparing various classification methods and finding the best method is beyond the scope of this study. Thus, we leave this for a future research topic.

5. Conclusion

For accurately classifying P300 signals, it is essential to thoroughly analyze and utilize temporal characteristics, such as latency, amplitude scale, and reverse. These temporal features play a pivotal role in understanding and interpreting P300 signals. Integrating these features effectively facilitates consistency across diverse datasets and enables effective classification. Hence, the SA method is a suitable approach for aligning and classifying P300 signals in cross-dataset environments.

In this study, we proposed a SA method for an ERP-based BCI. Four datasets were introduced to evaluate the SA methods, and we demonstrated that the SA works and results in improved performance in classifying EEG samples for the cross-paradigm datasets transfer learning concept compared with the standard approach.

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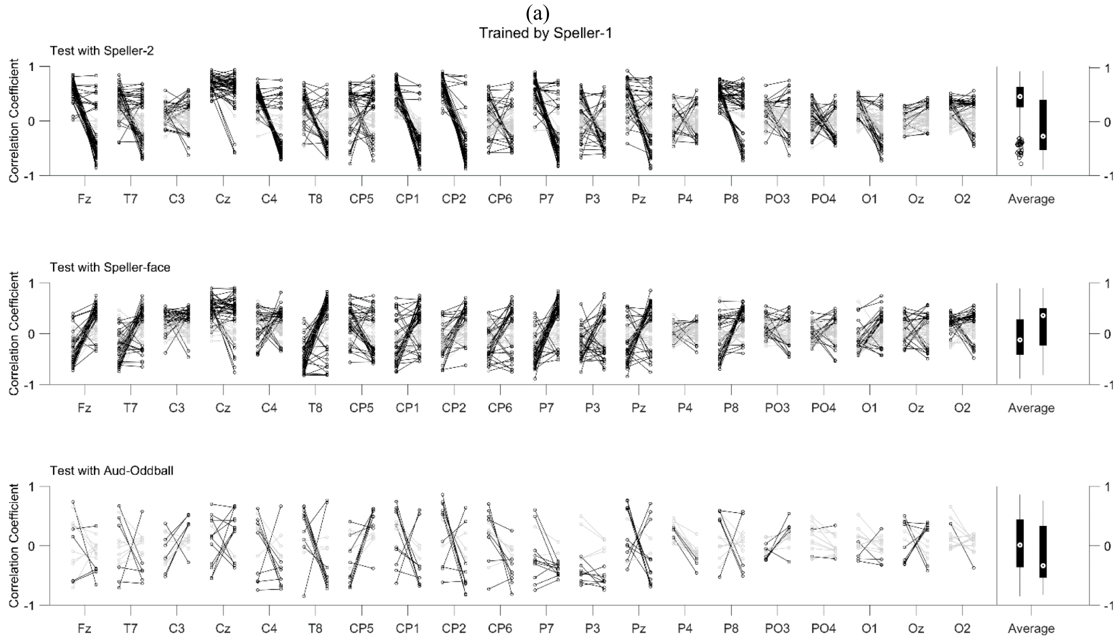
Data availability statement

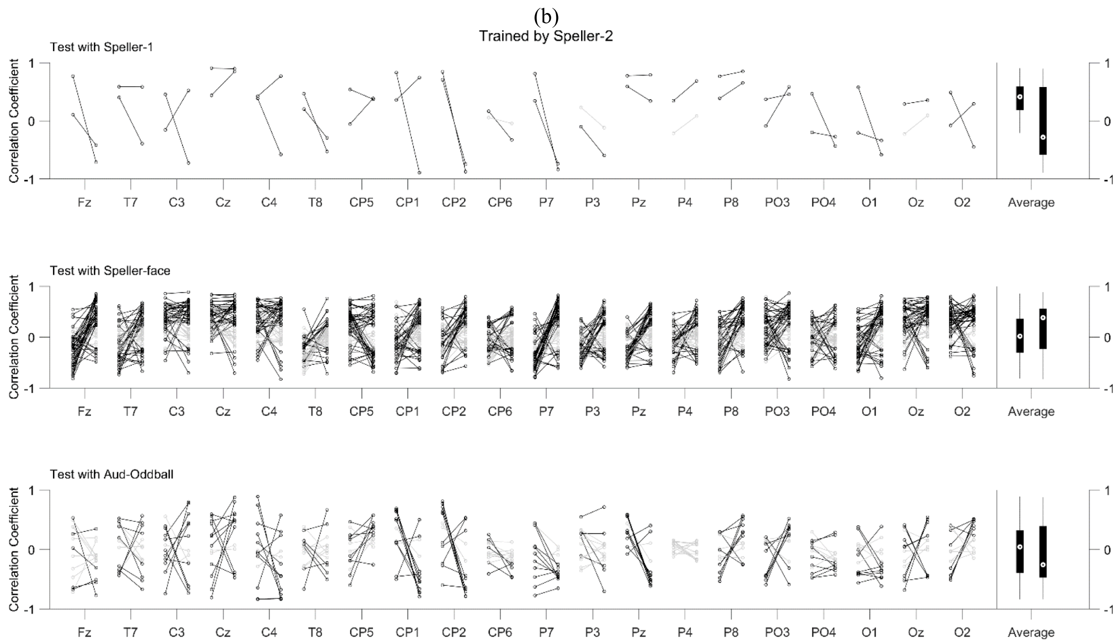
No new data were created or analysed in this study.

Appendix. Correlation coefficient values of standard (without alignment) and SA (with alignment)

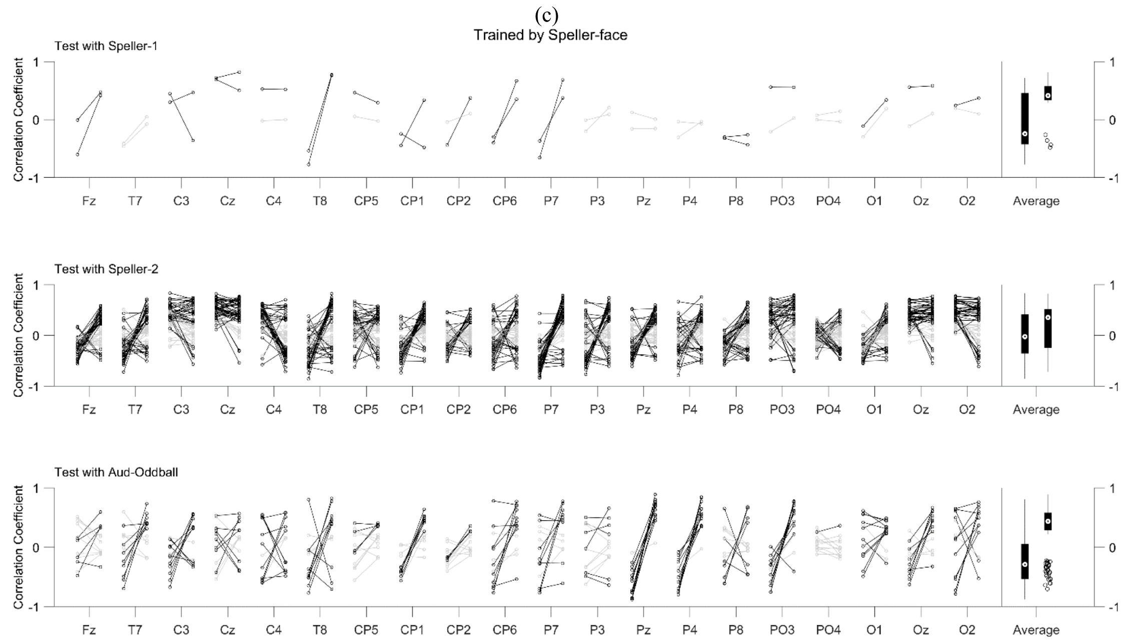
Correlation coefficient values when the SA method is and is not applied. The dots represent subjects, and the lines connect the correlation coefficient values with the standard method and SA method. Significant cases ($p < 0.05$) are marked with black lines and dots. The boxplot of the significant subjects is also shown on the right side of each figure.



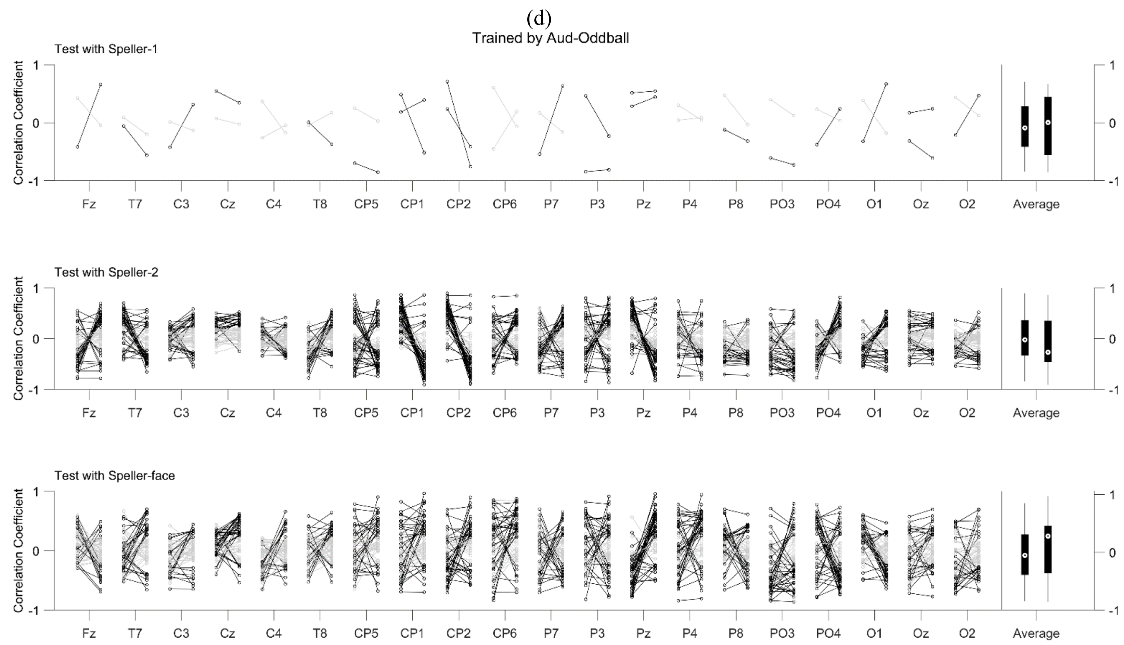
Appendix 1(a). Results of the model trained by Speller-1 dataset (from Speller-1 to other datasets).



Appendix 1(b). Results of the model trained by Speller-2 dataset (from Speller-2 to other datasets).



Appendix 1(c). Results of the model trained by the Speller-face dataset (from Speller-face to other datasets).



Appendix 1(d). Results of the model trained by the Aud-Oddball dataset (from Aud-Oddball to other datasets).

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